



* April 5, 2001

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Mattia Garatti

Curriculum Vitae

I am a Ph.D. student in Mathematics interested in Mathematical Analysis, and its interaction with Mathematical Physics and Geometry. In my master thesis I have worked on the variational representations of optimal transport and Schrödinger problem: the aim has been to highlight how the first can be recovered as Γ -limit of the second. Currently, I am carrying out my research on regularized optimal transport problem(s).

Education

Ph.D. in Mathematics, *Université Paris-Saclay*, Orsay, France. **2025 – ongoing**

Thesis: "On regularized Optimal Transport problems". Supervisors: Luca Nenna - Simona Rota-Nodari.

Master degree in Mathematics (LM-40), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2023 – 2025**

Average: 30 cum laude. Main courses' topics: Real and Functional Analysis, Measure Theory, Differential Geometry, Mathematical Physics. Thesis: "Variational representations of optimal transport and Schrödinger problem". Supervisor: Luca Tamanini - Cosupervisor: Nicolò De Ponti.

Bachelor degree in Mathematics (L-35), *Università Cattolica del Sacro Cuore*, Brescia, 110 cum laude. **2020 – 2023**

Average: 31.33/30. Thesis: "Il Teorema di esistenza di Peano per le equazioni differenziali ordinarie", provision of a theory on ODEs in the Peano's assumptions. Supervisor: Marco Marzocchi

High School Diploma, *Liceo Scientifico 'Camillo Golgi'*, Breno (BS), **2015 – 2020**
100/100.

Talks

Variational representations of optimal transport and Schrödinger problem, **2025**
Math PhD Days at UniTO, Turin.

Talk on my master thesis work

Teaching

MathExcellence 2025: corso di eccellenza in Matematica, *Università Cattolica del Sacro Cuore*, Brescia (BS). **June 2025**

Assistance and tutoring activities (5 hours)

Teacher Assistant, internship at *Liceo Scientifico 'Golgi'*, Breno (BS). **September 2022**

Frontal lectures for high school students, exam administration and correction, laboratory assistant

Private Teacher, internship at *Cooperativa 'Assolo'*, Berzo Inferiore (BS). **August – September 2018**

Lessons of Mathematics and Physics for high school students with SLD

Teacher Assistant, *supervision of preschool children, scientific experiments*, **June 2018**
Fondazione Scuola Materna 'San Giuseppe', Gratacasolo (BS).

Work experience

Math writer, Zanichelli editore S.p.A., Bologna (BO).

2024 – 2025

Teamwork with the editorial staff, autonomous creation of exercises of various difficulty on different topics to be included in future publications

Summer schools and workshop attended

Workshop in new Optimal Transport problems arising in various applications

Geometry, duality and convexity in new OT problems, *IMC Paris*

October 2025

BCMO Days 2025, *EDF Lab Paris-Saclay*, Palaiseau, France.

October 2025

Workshop on Optimal Trasport and its interactions with Data Sciences

Advances in Calculus of Variations and Nonlinear Analysis,

September 2025

Università Cattolica del Sacro Cuore, Brescia.

Workshop in honor of Marco Degiovanni's 70th birthday

Il Problema di Plateau, *Università Cattolica del Sacro Cuore*, Brescia.

April 2025

Lecture series about different approach to Plateau problem(s)

Introduction to Mechanics of Swelling, *Università Cattolica del*

February 2025

Sacro Cuore, Brescia.

Lecture series about mathematical models for swelling

Introduzione ai modelli matematici per i cristalli liquidi, *Università*

February 2024

Cattolica del Sacro Cuore, Brescia.

Lecture series about mathematical models for liquid crystals

Interuniversity Mathematics Summer School, *University of*

July – August 2023

Perugia.

Complex Analysis and PDE courses with related final exams successfully passed

Mathematics in Ljubljana, *University of Ljubljana, FMF and IMFM*.

July 2023

Summer School with courses on Formalization of Mathematics and Introduction to Complex Dynamics

Storia e tecniche dei metodi geometrici di rappresentazione:

March 2023

un'introduzione, *Università Cattolica del Sacro Cuore*, Brescia.

Mini-course about cartography and perspective

Un'Introduzione alla Teoria dei Grafi, *Università Cattolica del Sacro*

February 2023

Cuore, Brescia.

Mini-course about Graph Theory and its applications

Alcuni Aspetti della Matematica Antica, *Università Cattolica del Sacro*

March 2021

Cuore, Brescia.

Webinar about history of Mathematics and some greek geometry aspects

Corso di Eccellenza in Matematica, *Università Cattolica del Sacro Cuore*,

June 2019

Brescia.

Mini-course for high school students about different fields of Mathematics

Computer skills

Programming languages: Python, C, C++

Mark-up languages: LaTeX, HTML

Languages

Italian: Mother tongue

English: Professional proficiency

Awards	"Smart" study award , <i>Università Cattolica del Sacro Cuore</i> , Brescia (BS). 2025 Award for 100 best students of the university
	Merit scholarship , <i>Tenaris</i> , Dalmine (BG). 2017 – 2024 Scholarships awarded on merit every year during high school and university
	Merit scholarship , <i>Consorzio del Bacino Imbrifero Montano dell'Oglio</i> , Bergamo. 2021 – 2024 Scholarships awarded on merit every year during university
	Exemption from first year university fees , <i>Università Cattolica del Sacro Cuore</i> , Brescia. 2020 Exemption by merit
	Merit scholarship , <i>Comune di Costa Volpino</i> (BG). 2020 Scholarships awarded on merit for high school diploma
Organization	Member of the student jury of Galileo Prize. 2024 Jury for award regarding scientific dissemination
	Member of Joint Teachers – Students Commission , <i>Università Cattolica del Sacro Cuore</i> , Brescia. 2024 – 2025
	Staff member of Celebration of Mind, Notte dei Ricercatori, Venite a vedere la Terra che gira. Mathematical popularization days, <i>Università Cattolica del Sacro Cuore</i>
	Staff member of Disfida matematica. 2023 – ongoing Mathematical team competition for high school students, local phase in Brescia
	Member of Ufficio di Sezione Elettorale , <i>Comune di Costa Volpino</i> (BG). 2021 – ongoing Scrutineer and president of polling station
	Parish volunteer , <i>Parrocchia di "San Zenone"</i> , Gratacasolo. 2015 – ongoing Organization of summer camps and member of the local choir with budget manager functions

I hereby authorize the use of my personal data in accordance to the GDPR 679/16 - "European regulation on the protection of personal data".

November 24th, 2025

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